

Beyond Individual AI Literacy: Family AI Literacy as Cooperative Accomplishment

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Families encounter artificial intelligence (AI) through shared learning, caregiving, safety, and decision-making, yet AI literacy frameworks usually represent competence as individual. We conducted a two-layer secondary content analysis of 397 unique AI/GenAI literacy framework papers drawn from a systematic review; 393 met the eligibility criteria for this analysis. Of these, 379 (96.4%) located literacy primarily in individuals, 308 (78.4%) encoded no structurally necessary human relationship, and only one (0.3%) treated a family or household as the literacy unit. A second-stage analysis examined cooperative processes in 98 papers meeting relational or family criteria. Coordination and collective agency each appeared in 67 papers (68.4%), and 50 (51.0%) combined joint interpretation, collective governance, and coordinated action. Yet 38 of those 50 still assigned literacy primarily to individuals. This mismatch supports a definition of family AI literacy as the situated capacity to make AI understandable, governable, and actionable together. The codebook and three-function account distinguish AI located in family settings from AI organized around family relationships and make visible whose interpretation, authority, labor, and action count.

Additional Key Words and Phrases: AI literacy, generative AI, family-centered AI, families, cooperative work, relationality, articulation work

1 Introduction

AI enters family life through schoolwork, shared devices, caregiving, health, privacy, and safety. These encounters are interdependent: one person prompts while another checks; a child introduces a tool that a parent governs; a caregiver verifies consequential advice. Research on AI in family life documents changing routines, roles, safety practices, learning relationships, and intergenerational oversight [1–4, 9–11]. Yet influential AI-literacy accounts emphasize individual recognition, application, evaluation, and ethics [6, 7]. Whether the broader framework literature represents joint interpretation, verification, rule negotiation, and responsibility remains unclear.

Using corpus-wide secondary content analysis, we ask: (RQ1) where frameworks locate literacy’s holder and enactment; (RQ2) how they position families and intergenerational relationships; and (RQ3) which cooperative processes they encode. We treat family AI literacy as one evaluative dimension of family-centered AI: whether a system organizes competence as an individual possession or supports cooperative interpretation, governance, and action. We contribute a corpus-wide account of the mismatch between cooperative processes and individual literacy ownership, a two-layer analytic vocabulary, and an interpretive definition traced to recurring configurations of joint interpretation, collective governance, and coordinated action.

2 Method

2.1 Corpus and eligibility

The source systematic review retrieved 801 full-text records. Eligible English-language papers published from 2018 to 2026 contributed an original, codeable AI/GenAI literacy framework, model, taxonomy,

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curriculum, intervention, or instrument; applications, reviews without new synthesis, and adjacent constructs were excluded. Of 442 eligible papers, 400 met quality criteria for structural validity, sample adequacy, or transparent derivation. After three duplicates were removed, 397 papers were coded; four fell outside the review’s AI-literacy definition, leaving 393 for prevalence analysis.

2.2 Two-layer codebook and verification

Coding followed the two-layer codebook summarized in Table 1. Evidence came from each paper’s focal artifact. Material from literature reviews, participant descriptions, development activities, and cited frameworks counted when incorporated into that artifact. Layer 1 separated literacy holder, relational necessity, family role, and structural centrality, so a family mention did not by itself establish family-held literacy. Family evidence included explicit family, household, caregiver, parent, and intergenerational relations; a family locus was assigned only when the focal artifact made that collective the holder or necessary enactor of literacy.

Table 1. Reader-facing summary of the two-layer codebook.

Decision	Codes	Reader-facing rule
Literacy locus and enactment	IND, DYAD, FAM, GRP, ORG, SOC, MULTI; R0–R3	Identify the primary holder of literacy and whether interaction or a collective unit is required.
Family role and centrality	F0–F7/FU; NA, S0–S3	Distinguish family absence, context, participation, reporting, scaffolding, co-learning, and family-held literacy; then locate that evidence from background to core definition.
Layer 2 entry	F2–F7/FU; R2/R3; FAM/GRP/MULTI; or S2/S3	Review cooperative processes when any family, relational, collective, or structurally central criterion is met.
Cooperative processes	Coordination; sensemaking; shared judgment; norm negotiation; reciprocal scaffolding; collective agency; distributed responsibility; authority and roles	Code each process independently from direct evidence in the focal artifact; an emergent residual captures processes outside these eight.
Process status	P0, P1, P2	Record a reviewed absence, a present non-core process, or a named, repeated, or structurally central process.

A paper entered Layer 2 if it met any criterion: F2–F7/FU family position; R2/R3 relational status; FAM, GRP, or MULTI locus; or S2/S3 centrality. The 98 qualifying papers were reviewed for eight processes and an emergent residual. P1 and P2 were analyzed as present; each required direct evidence. P0 followed review of the relevant dimensions, items, or mechanisms.

The 30-paper pilot included all 12 family-term papers and 18 contrastive cases. Reviews every 50 papers clarified family terminology, human versus virtual peers, and participatory development versus participation in the artifact, and prompted recoding where needed. AI-supported PDF search located candidate passages; the author selected evidence and assigned all codes. Item-level examination changed ten codes whose dimension labels obscured interpersonal items.

2.3 Analysis

Layer 1 percentages use 393 papers; Layer 2 percentages use 98. Cross-artifact comparisons are descriptive; sensitivity analysis compared 397 coded papers with 393 eligible papers.

After descriptive coding, we grouped processes by cooperative function: sensemaking and shared judgment as mutual intelligibility; norm negotiation, authority and roles, and distributed responsibility as collective

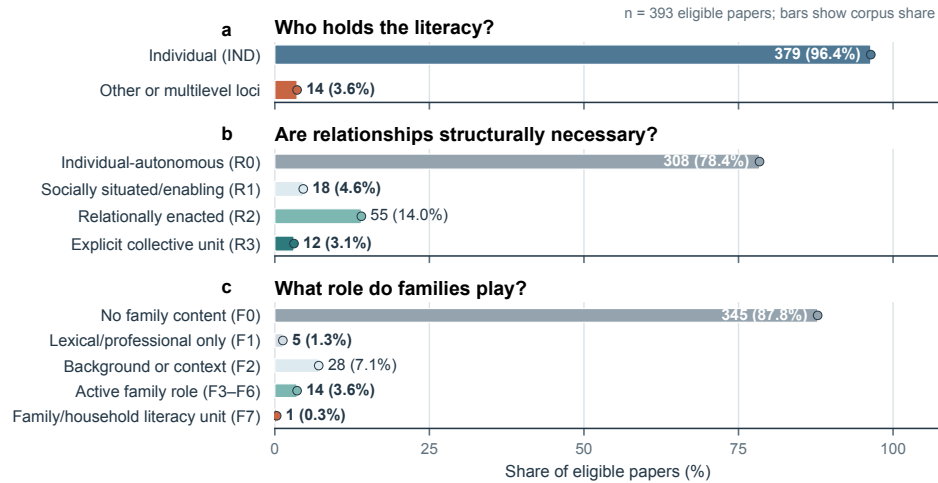
AI-literacy frameworks overwhelmingly center individuals, not relationships or families

Fig. 1. Corpus orientation in the 393-paper eligible analysis set. (a) Primary locus of literacy. (b) Relational status. (c) Family orientation. Bars report corpus shares and labels give counts and percentages; percentages may not sum to 100 because of rounding.

governance; and coordination, reciprocal scaffolding, and collective agency as coordinated action. Cluster presence and co-occurrence were compared with literacy locus, family role, high-centrality cases, and negative cases. This sequence links each term in the definition to specified codes and configurations.

3 Results

3.1 Individual competence dominates the corpus

AI literacy was overwhelmingly assigned to individuals (Figure 1a–c). Of the 393 eligible papers, 379 (96.4%) used an individual primary locus; the remaining 14 used linked multilevel (10), group (2), organizational (1), or family (1) loci. Likewise, 308 papers (78.4%) encoded no structurally necessary human relationship. Eighteen (4.6%) placed an individual in a social context without requiring joint enactment, 55 (14.0%) required an interpersonal relationship, and 12 (3.1%) defined an explicit collective unit.

Family evidence was even less central. Most papers contained no family content (345/393, 87.8%); five (1.3%) contained only lexical or professional uses, and 28 (7.1%) placed family in background or context. Fourteen papers (3.6%) assigned family members an active role as a target, proxy reporter, mediator, scaffold, or co-learner. Only one paper, the 4As, treated the family or household as the holder or necessary enactor of AI literacy (0.3%).

3.2 Cooperative processes exist without family-held literacy

Ninety-eight papers (24.9%) were included in Layer 2. Coordination and collective agency were each present in 67 papers (68.4%), authority and role dynamics in 59 (60.2%), reciprocal scaffolding in 57 (58.2%), and sensemaking in 55 (56.1%). Shared judgment appeared in 45 papers (45.9%). Distributed responsibility (34/98, 34.7%) and norm negotiation (28/98, 28.6%) were less common.

At the level of functional clusters, mutual intelligibility appeared in 61 papers (62.2%), collective governance in 65 (66.3%), and coordinated action in 78 (79.6%). Fifty papers (51.0%) contained at least one process from all three clusters. Of these, 38 retained an individual primary locus and only one used a family locus. Seventeen Layer 2 papers contained no positive process after full review; 12 of them placed family only in background or context. These contrasts separate family mention from cooperative enactment and expose the structural mismatch between the processes encoded and the unit that holds literacy.

4 Discussion: Family AI Literacy as Cooperative Accomplishment

Family members usually appeared as context, participants, reporters, or supporters of another person’s competence. Meanwhile, the three functional clusters appeared together in half of the Layer 2 papers, but most of those papers retained an individual literacy locus. Family presence, cooperative enactment, and family-held literacy are therefore distinct conditions.

We therefore define *family AI literacy* as a cooperative accomplishment: the situated capacity to make AI understandable, governable, and actionable together. *Understandable* means that members can surface interpretations, question outputs, and establish a shared basis for judgment. *Governable* means that permissions, roles, authority, and responsibility can be allocated and revised. *Actionable* means that members can coordinate participation, scaffold one another, and form or carry out a joint decision. *Together* locates the capacity in the organization of interdependent contributions while allowing unequal knowledge, differentiated authority, and continuing disagreement.

This interactional account brings literacy as social practice [5] into conversation with research on articulation work, which explains how people align tasks, responsibilities, and dependencies [8]. It also foregrounds the unequal authority and interdependence that characterize family care.

Because family mention usually left literacy individual, family-centered AI cannot be identified by a household setting, a parent-child example, or several users. It should let participants define relevant relations of care, kinship, and interdependence and make visible whose knowledge, authority, and labor sustain AI across changing family and institutional contexts.

Because coordination (68.4%) far exceeded norm negotiation (28.6%) and distributed responsibility (34.7%), evaluation should ask not only whether family members coordinate, but who verifies, mediates, and troubleshoots and whether authority matches responsibility. Designs can support visible handoffs, revisable permissions, preserved disagreement, explicit accountability, and role-sensitive controls.

The evidence characterizes how framework documents represent literacy. Empirical studies with families are needed to examine how these processes vary across homes and settings.

For the workshop, the codebook offers a boundary object across learning, care, and home-institution settings. Three questions organize comparison: Whose interpretation must become shareable? Who can revise permissions, roles, and responsibility? What joint action should the system enable, and who carries the work? These distinguish family-located from family-organized AI and provide a concrete starting point for the workshop’s definition and design agenda.

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